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ER Log, LV Log, C RLV Writing Journal (3)

Reading Log

Component	Entry
Entry #	R005
Date	November 10, 2025
Title	Accelerating AI Inference on Edge Devices: A Hardware Perspective
Source / Website	ACM Digital Library (2023)
Text Type	Research Article / Hardware Engineering
Text Level	C1
Topic / Theme	Edge AI, TPU, FPGA, Latency
Length (In Page)	11 pages
Difficulty (1–5)	4/5
New Vocabulary	Throughput, Quantization, Neural Engine, Hardware Accelerator
Key Career Insights	Optimizing AI models for hardware constraints is a key skill for developing responsive mobile and embedded systems.
Recommendation	★★★★☆

Component	Entry
Entry #	R006
Date	November 15, 2025
Title	Analysis of I/O Scheduling Algorithms in the Linux Kernel
Source / Website	Linux Journal / Open Access (2022)
Text Type	Technical Article / Operating Systems
Text Level	B2+
Topic / Theme	Linux Kernel, I/O Scheduling, Performance Tuning
Length (In Page)	9 pages
Difficulty (1–5)	4/5
New Vocabulary	Throughput, Deadlock, Latency, Block Device
Key Career Insights	Deep knowledge of the Linux kernel allows engineers to tune system performance for specific high-load applications.
Recommendation	★★★★☆

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Reading–Writing Journal

RWJ Entry 5

C1 Level (Week 3 of 8)

Writing Prompt Response: “Hardware Optimization for Artificial Intelligence”

Entry: 3/8 – Week 3

My Journal Entry:

In Entry R005, I explored how AI models are optimized to run on low-power edge hardware like the Jetson Nano or specialized TPUs. The paper explains "Quantization," a process that reduces the precision of model weights to save memory and speed up computation. I found it fascinating how hardware accelerators can drastically reduce latency compared to standard CPUs. This is very practical information for working with advanced models on portable devices. Understanding the trade-offs between accuracy and hardware throughput is essential for building real-time applications that don't require expensive cloud servers, which is vital for the future of mobile technology.

Word Count: 117

RWJ Entry 6

B2+ Level (Week 3 of 8)

Writing Prompt Response “Linux System Performance and Management”

Entry: 3/8 – Week 3

My Journal Entry:

In Entry 2006, this article explains how the Linux kernel manages input/output (I/O) operations between the processor and storage devices. It compares different scheduling algorithms like CFQ, Deadline, and NOOP. I learned that choosing the right scheduler can significantly impact the performance of a server or an embedded system. For instance, some algorithms are better for solid-state drives while others favor traditional hard disks. This technical reading deepens my understanding of system administration and performance optimization. It is very useful for managing Linux-based environments where high-speed data processing is required, allowing for better resource management in complex computing infrastructures.

Word Count: 113